

Stage 6 Integrated Workflow Report

Claim Synthesis and Final Paper Preparation

Codex-assisted integrated workflow review

June 21, 2026

Abstract

This report synthesizes the completed solver validation, v1 small-grid dataset, price/premium surrogate, boundary diagnostic component, and Delta diagnostic component. It does not train a new model, regenerate the dataset, introduce a Gamma head, or modify solver code. The Stage 6 review decision is READY_FOR_FINAL_PAPER_DRAFTING.

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1 Purpose of Stage 6

Stage 6 is an integrated review and final-paper preparation pass. Its job is to connect each technical component to a claim that can safely appear in a draft paper. The methodology guides emphasize that American option risk-surface work should begin with a validated finite-difference benchmark and only then add constrained, diagnostic neural components (Brennan and Schwartz, 1977; Wilmott et al., 1995).

2 Compressed Roadmap Status

The compressed roadmap has completed five stages before this synthesis: v0 dry run, v1 dataset QA, price/premium surrogate comparison, boundary diagnostics, and Delta diagnostics. Stage 6 does not add a sixth model; it audits the evidence produced by those stages.

3 Component Evidence Summary

Component	Method	Key Metric	Status
CN/PSOR solver benchmark	Baseline American Crank-Nicolson / PSOR LCP solver	PASS_SOLVER_VALIDATION_	READY_WITH_LIMITATIONS
v1 small-grid dataset	288 approved regimes sampled on S/K in [0.4,1.8]	288 accepted regimes; 1498464 rows; max obstacle 0.0	READY_WITH_LIMITATIONS
price/premium surrogate	Positive-premium MLP with analytic payoff reconstruction	test value RMSE 0.012269; stress RMSE 0.009516; test obstacle rate 0.000e+00	READY_WITH_LIMITATIONS
boundary diagnostic component	Direct boundary-head diagnostic baseline	test RMSE 0.041007; stress RMSE 0.084293	READY_WITH_LIMITATIONS
Delta diagnostic component	Supervised bounded Delta-head diagnostic baseline	test RMSE 0.057626; stress RMSE 0.048061; test bounds rate 0.000e+00	READY_WITH_LIMITATIONS
Gamma component	No Gamma head trained	Gamma remains blocked	BLOCKED

4 Integrated Workflow Description

The retained workflow is modular. A validated CN/PSOR benchmark generates diagnostic surfaces. The v1 small-grid dataset records regime-level splits, labels, masks, and diagnostics. The positive-premium model handles price and continuation premium. A direct boundary-head diagnostic component handles threshold-based free-boundary labels. A supervised Delta-head diagnostic component handles finite-difference Delta labels. Gamma remains blocked rather than silently converted into a model claim.

5 Headline Metrics

Component	Metric Name	Split Or Region	Value
dataset	accepted_regimes	v1_small_grid	288
dataset	accepted_rows	v1_small_grid	1498464
dataset	max_obstacle_violation	v1_small_grid	0.0
price/premium	positive_premium_value_rmse	test	0.012268641982174276
price/premium	positive_premium_value_rmse	stress_holdout	0.009515969641226743
boundary	direct_boundary_head_rmse	test	0.04100654163466295
boundary	direct_boundary_head_rmse	stress_holdout	0.08429296961550176
Delta	supervised_delta_head_rmse	test	0.05762588026155442
Delta	supervised_delta_head_rmse	stress_holdout	0.04806100815073663
Gamma	gamma_head_status	overall	BLOCKED

6 Price/Premium Component Synthesis

The Stage 3 evidence supports the positive-premium representation. Predicting nonnegative continuation premium and reconstructing value with analytic payoff aligns the surrogate with the American obstacle. This supports a price and premium surface claim, but not a boundary or Greek claim by itself.

7 Boundary Component Synthesis

The Stage 4 evidence is deliberately two-part. The premium-implied boundary method is not reliable in this test because many rows have no comparable boundary-error points. The direct boundary-head diagnostic baseline satisfies the boundary diagnostics and supports a separate boundary-focused component. The boundary remains a threshold-based CN/PSOR diagnostic, not exact analytical truth.

8 Delta Component Synthesis

The Stage 5 evidence shows that good price or premium accuracy does not automatically imply production-quality Delta. Price-implied Delta diagnostics are informative, but the supervised Delta head is the retained diagnostic component because it satisfies the RMSE and bounds/sign criteria. Delta labels remain finite-difference diagnostics.

9 Blocked Gamma and Production Claims

No Gamma head is trained in the compressed roadmap. Production pricing, exact free-boundary, production Greek, broad extrapolation, and one-model-solves-all claims are blocked. This is a strength of the synthesis: weak claims are removed rather than hidden.

10 Claim Evidence Matrix

Claim Id	Claim Text	Status	Final Paper Use
C01	A validated baseline CN/PSOR benchmark can generate consistent American-option diagnostic surfaces within the approved one-dimensional Black-Scholes grid.	SUPPORTED_WITH_LIMITATIONS	dataset construction sections.
C02	Positive continuation-premium prediction is better aligned with the American payoff obstacle than direct value prediction.	SUPPORTED_WITH_LIMITATIONS	price premium surrogate section.
C03	Boundary behavior should be handled by a boundary-focused diagnostic component rather than only thresholding price/premium predictions.	SUPPORTED_WITH_LIMITATIONS	boundary diagnostic component section.
C04	Delta behavior should be handled with a Delta-focused diagnostic component rather than assumed from price accuracy.	SUPPORTED_WITH_LIMITATIONS	Delta focus component section.
C05	The project contribution is a modular research workflow, not a single universal neural solver.	SUPPORTED_WITH_LIMITATIONS	integrated workflow and conclusion sections.

11 Blocked Claims Matrix

Blocked Claim	Reason Blocked	Possible Future Work
Production pricing or production risk-system readiness	The workflow is validated as a research diagnostic pipeline only.	Add calibration, monitoring, runtime SLAs, and independent production QA.
Exact analytical free-boundary accuracy	Boundary labels are threshold-based CN/PSOR diagnostics.	Compare against high-resolution references and alternative boundary methods.
Production Greek reliability	Delta is a finite-difference diagnostic label and Gamma remains fragile.	Add higher-grid Greek confirmation, hedging diagnostics, and uncertainty checks.
Gamma-head or Gamma-surrogate claims	No Gamma head is trained in the compressed roadmap.	Design a separate masked Gamma diagnostic stage after human review.
Broad extrapolation outside the approved v1 grid	The dataset uses a fixed small Cartesian grid and regime-level splits.	Run a larger-regime envelope study with PSOR spot checks.
One surrogate model solves price, boundary, and Greeks together	Stages 4 and 5 show separate boundary and Delta components are needed.	Study integrated multi-head architectures only after preserving component diagnostics.

12 Final Paper Outline

Section Number	Section Title	Purpose	Key Evidence
1	Introduction	Motivate American option risk surfaces and the free-boundary challenge.	Planning report and methodology guide.
2	Research question	State the modular structure-preserving surrogate question.	Application strength gate and Stage 6 claim matrix.
3	Background and American option formulation	Define payoff obstacle, LCP, boundary, Delta, and Gamma caveats.	Formulation note and references.
4	Numerical benchmark: CN/PSOR and LCP validation	Document the validated finite-difference benchmark.	Ticket 12 synthesis and v1 QA.
5	Dataset construction	Explain v1 regimes, splits, masks, and diagnostics.	v1 small-grid report.
6	Price / premium surrogate	Show why positive premium is the retained price component.	Stage 3 metrics and obstacle summary.
7	Boundary diagnostic component	Explain why direct boundary diagnostics are needed.	Stage 4 boundary metrics.
8	Delta diagnostic component	Explain why Delta needs a supervised diagnostic head.	Stage 5 metrics and bounds summary.
9	Integrated workflow	Assemble the component workflow and claim matrix.	Stage 6 readiness matrices.
10	Limitations	State blocked claims and scope boundaries.	Blocked claims matrix.
11	Conclusion	Summarize the cautious research contribution.	Supported claim matrix.
12	Future work	Describe larger regimes, active learning, Gamma diagnostics, and production gaps.	Handouts and blocked claims.

13 Figure and Table Inventory

Item Id	Item Type	Recommended Final Paper Section	Purpose
F01	figure	Price / premium surrogate	Compare direct value and positive-premium held-out errors.
F02	figure	Price / premium surrogate	Show obstacle and negative-premium behavior.
F03	figure	Boundary diagnostic component	Show direct boundary-head diagnostics and premium-implied limitations.
F04	figure	Boundary diagnostic component	Show representative put boundary curves.
F05	figure	Delta diagnostic component	Show Delta error across held-out splits.

Item Id	Item Type	Recommended Final Paper Section	Purpose
F06	figure	Delta diagnostic component	Show bounds/sign diagnostics.
T01	table	Integrated workflow	Summarize component readiness.
T02	table	Integrated workflow	Map final claims to evidence.

14 Stage 6 Figures

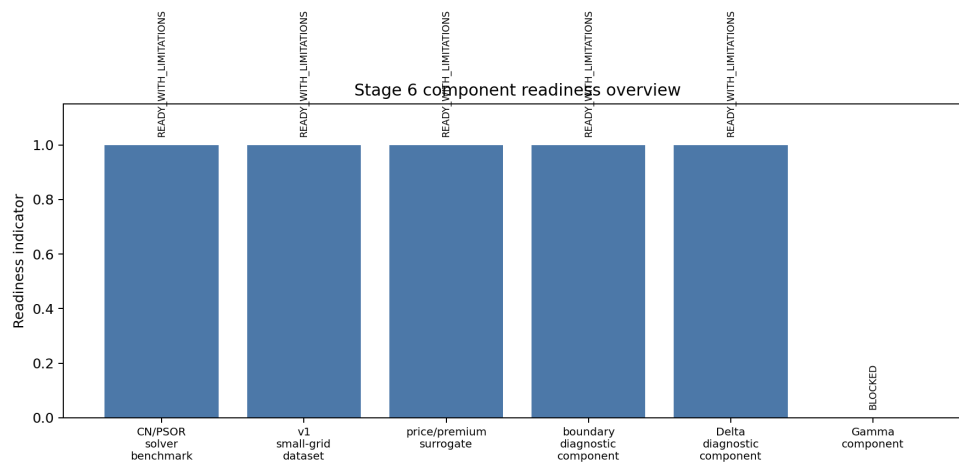


Figure 1: Component readiness overview.

Integrated evidence map: modular workflow, Gamma blocked

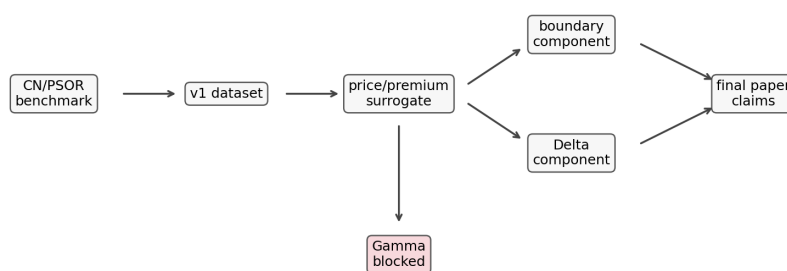


Figure 2: Workflow evidence map.

15 Limitations

- The integrated workflow is a synthesis of existing evidence, not a fresh PSOR benchmark run.
- The v1 dataset is limited to the approved small-grid Black-Scholes regime envelope.
- Boundary labels are threshold-based diagnostics.

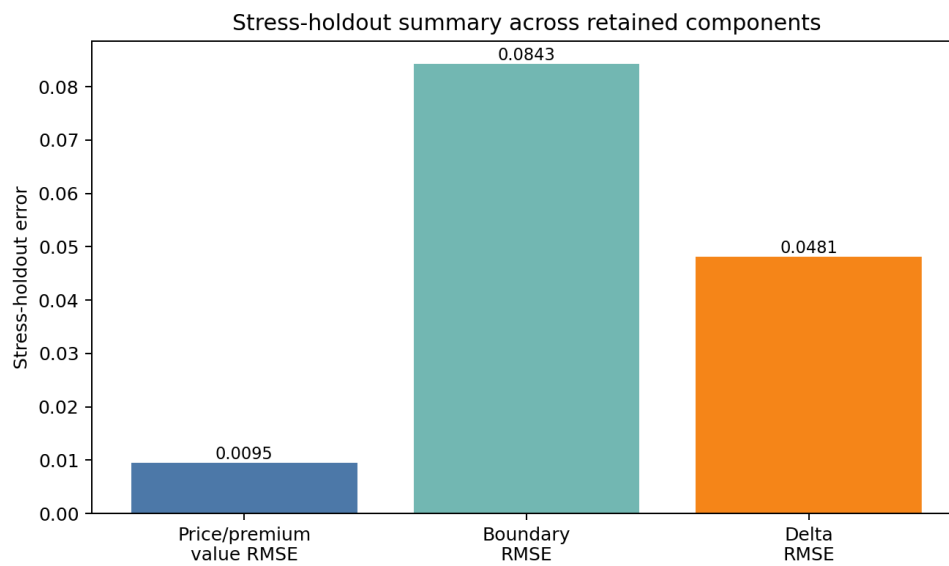


Figure 3: Stress-holdout summary across retained components.

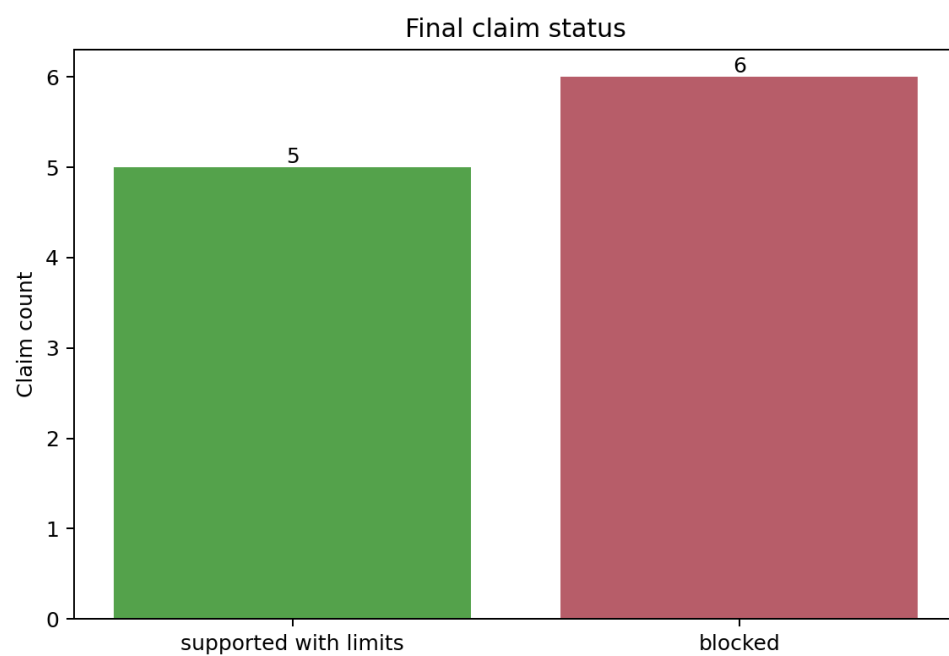


Figure 4: Final supported and blocked claim counts.

- Delta labels are finite-difference diagnostics.
- Gamma is blocked from surrogate claims.
- The final paper still needs careful prose, citation integration, and human review.

16 Final Readiness Decision

READY_FOR_FINAL_PAPER_DRAFTING

The evidence is strong enough to begin final paper drafting, provided the paper keeps the modular claim structure and the blocked-claims language intact.

17 Recommended Next Step

Draft the final paper from `reports/07_integrated/final_paper_outline.md` and the Stage 6 matrices. Do not add a Gamma claim, production claim, or universal neural-superiority claim unless new evidence is generated in a separate approved stage.

References

- Michael J. Brennan and Eduardo S. Schwartz. The valuation of american put options. *The Journal of Finance*, 32(2):449, 1977. doi: 10.2307/2326779.
- Paul Wilmott, Sam Howison, and Jeff Dewynne. *The Mathematics of Financial Derivatives*. Cambridge University Press, 1995. doi: 10.1017/CBO9780511812545.